MIL-G-46693F(EA)
23 March 1976
SUPERSEDING
MIL-G-46693E(MU)
16 October 1968

MILITARY SPECIFICATION

GRENADE, HAND-RIFLE, SMOKE, WP, M34; INERT COMPONENTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inert components for one type grenade.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-B-7883 - Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.

STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 1330

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DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

D13-7-4 - Grenade, Hand-Rifle, Smoke, WP, M34.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

R. M. GRAZIANO TARIFF No. 29

(Issued by R. M. Graziano, Agent, 1920 L Street, N.W. Washington, DC 20036).

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D13-7-4 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the grenade shall conform to the specifications and drawings listed on Drawing D13-7-4 and subsidiary drawings.

3.2 Assembly. All subassemblies of the grenade shall conform to their respective drawings.

3.3 Brazing. The body and cap assembly, and the burster well assembly shall be brazed in accordance with MIL-B-7883 Grade B and tested as specified in 4.4.4.3.

3.4 Body and cap, burster well leakage. When tested in accordance with 4.4.4.1, neither assembly shall leak while each is subjected to an internal air pressure of 45 $\pm$ 5 pounds per square inch gage (psig) for a period of 35 $\pm$ 3 seconds after the 45 $\pm$ 5 psig pressure is reached.

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3.5 Hardness. Body hardness shall be as specified on Drawing C13-7-10 when tested as specified in 4.4.4.2.

3.6 Preproduction. Prior to the start of regular production, a preproduction lot of components shall be produced in accordance with this specification for examination and tests (see 4.3).

3.7 Workmanship. The components of the grenade shall be free from foreign matter (oil, grease or liquid) and damage such as dents, cracks or deformation which could affect their intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 200 each body and cap assemblies and burster well assemblies shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

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4.3.2 Sampling.

4.3.2.1 For examination and nondestructive tests. Twenty-five of each assembly shall be randomly taken from the preproduction lot.

4.3.2.2 For destructive test. Ten body and cap assemblies shall be randomly taken from the preproduction lot.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive tests. The sample assemblies shall be examined and tested in accordance with the classification of defects (4.4.3.4). The AQL's do not apply.

4.3.3.2 For destructive test. The sample assemblies shall be tested in accordance with 4.4.4.3.

4.3.4 Acceptance/rejection criteria. The preproduction samples shall comply with the examinations and tests specified in 4.3.3 to be acceptable. The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of like components produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-3.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample assemblies shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive test. The sample assemblies shall be tested in accordance with 4.4.4.3 using an AQL of 1.0.

4.4.3.3 For critical defects. Each item in the lot shall be tested for its critical characteristic listed in the classification of defects.

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4.4.3.4 Classification of defects.

(a) Body and cap assembly (Dwg C13-7-10).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.4.4.1(a)
Major: AQL 0.65 percent defective		
101	Brazing	
AQL 1.0 percent defective		
102	Overall length incorrect	
103	Press-fit diameter or taper incorrect	
104	Machine finish (16-32) incorrect	
105	Neck profile to 1.735 length incorrect	
106	Workmanship (3.7)	
Minor: AQL 2.5 percent defective		
201	Corrosion preventive missing	

(b) Well, burster assembly (Dwg C13-20-82).

Categories	DefectsAcceptance standards	
Critical:		
1	Leakage	4.4.4.1(b)
Major: AQL 0.65 percent defective		
101	Brazing	
AQL 1.0 percent defective		
102	Thread incorrect	
103	Press-fit diameter incorrect	

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104	Machine finish (16-32) incorrect
105	Eccentricity excessive
106	Workmanship (3.7)

Minor: AQL 2.5 percent defective

201	Radius (press-fit surface) missing
202	Corrosion preventive missing
203	Overall length incorrect

4.4.4 Tests.

4.4.4.1 Leakage.

(a) Body and cap assembly (Drawing C13-7-10). The body and cap assembly leakage shall be determined by using Tester, Air Leakage, Q48.

(b) Burster well assembly (Drawing C13-20-82). After completion of all operations on the burster well assembly, leakage shall be determined by using Tester, Air Leakage, Q48.

4.4.4.2 Hardness. The body shall be sectionalized longitudinally to obtain a 1-inch wide strip. The hardness shall be taken on the approximate center line of the strip at three places as follows:

(a) Approximately 1 inch from each end.
(b) Approximately in the center of the strip.

One or more indications on any one strip that the positions do not conform to the required hardness shall constitute one defect.

4.4.4.3 Brazing. The brazing shall be visually examined in accordance with MIL-B-7883.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking, interplant shipment. The components shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further prccessing. Shipping containers shall be in compliance with rules and regulations applicable to the mode of transportation and R. M. Graziano's Tariff, as applicable. Marking shall be in conformance with FED-STD-123 and shall include the contractor's name, address, and contract number.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction.

(1) Time allowed for suppliers submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1330-A117

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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